**Maxime RYCKEWAERT**
36 years old
Cirad/GSP team

# RESUME

## OVERVIEW

I am a researcher in artificial intelligence and machine learning, focusing on the analysis of complex multivariate and multimodal data. My work develops advanced methods to extract meaningful information from high-dimensional signals, with applications in agriculture, environmental sciences, biodiversity, and ecology to support sustainable decision-making.

## EDUCATION AND DEGREES

2016-2019: **PhD Cifre**, Limagrain, Irstea, Institut Agro Montpellier
2011-2013: **Master's Degree** in Physics (Computer Physics), University of Montpellier
2008-2011: **Bachelor's Degree** in Fundamental Physics, University of Montpellier

## PROFESSIONAL EXPERIENCE

04/2025: **Permanent Researcher,** Cirad, UMR AGAP, GSP Team (Montpellier)
Currently holding a permanent position in the *Genome and Selection of Perennial Crops* (GSP) team, my research develops **artificial intelligence** methods to **support sustainable plant breeding in perennial crop**s such as forests, vineyards, cacao, and coffee. More broadly, my work addresses **the tensions between agriculture, biodiversity conservation**, and ecological sustainability by integrating genomics, phenotyping, and species distribution modeling. Using AI-driven approaches at large spatial scales, including continental analyses, I aim to inform **land-use management and conservation strategies** that balance agricultural production with long-term biodiversity preservation.

09/2023-03/2025: **Starting Research Position**, Inria, UMR LIRMM, Zenith Team (Montpellier)
**Project**: Horizon Europe B-CUBED: Biodiversity Building Blocks for Policy
**Responsibilities**: Development of deep-learning models using biodiversity data cubes to model species distribution across Europe. Focus on integrating environmental data, remote sensing, and human footprint data to reduce biases from heterogeneous citizen science sampling efforts.

08/2022-07/2023: **Postdoctoral Researcher**, INRAE, UMR ITAP, COMiC Team (Montpellier)
**Project**: Casdar BEETLENIRS
**Topic**: Detection of grain pest insects using hyperspectral imaging
Responsibilities: INRAE lead for the development of a new hyperspectral imager and its associated data processing workflows.

11/2020-07/2022: **Postdoctoral Researcher,** INRAE, UMR ITAP, COMiC Team (Montpellier)
Project: Interreg Sudoe VINIOT
**Topic**: Detection of water stress, disease identification, and maturity monitoring of vines through hyperspectral imaging
**Responsibilities**: INRAE lead for a work package (scientific coordination, experimentation, data analysis, and dissemination).

11/2019-10/2020: **Postdoctoral Researcher**, INRAE, UMR ITAP, COMiC Team (Montpellier)
**Project**: ANR OPTIPAG
**Topic**: Optical instruments for precision agriculture: sunflower phenotyping using hyperspectral imaging for varietal selection
**Responsibilities**: Big data analysis, speckle imaging, and supervision of junior researchers.

11/2016-10/2019: **PhD Candidate** Cifre, Limagrain (Clermont-Ferrand)
**Topic**: Potential of coupling a high spectral resolution/low spatial resolution sensor with a low spectral resolution/high spatial resolution sensor for varietal selection: application to maize under water stress conditions.

## SKILLS

**Data Analysis**: Machine learning methods, neural networks, point processes, multivariate statistical data analysis, Bayesian statistics, unsupervised/supervised methods (exploration, discrimination, regression, classification), multi-block methods, robust methods, preprocessing, variance analysis, pan-sharpening, local methods, variable selection, dimensionality reduction, signal processing, multi- and hyperspectral image processing.

**Mathematics**: Linear algebra, numerical analysis, complex analysis, numerical optimization, …

**Physics/Digital Physics**: Electromagnetism, optics, statistical physics, …

**Experimentation**: Hyperspectral imaging (VIS-NIR, SWIR), multispectral imaging, spectrometry (FTIR, VIS, NIR, SWIR), speckle and biospeckle imaging.

**Programming**: Python, PyTorch, R, MATLAB, C++/C, Java; High-Performance Computing (HPC): GPU, MPI, OpenMP; LaTeX.

## LANGUAGES

French and English

## INTERNATIONAL EXPERIENCE

**2021**: University of L'Aquila, Italy, Explore mobility project, MUSE
**2021**: University of Sapienza, Rome, Italy, Explore mobility project, MUSE
**2018**: University of Sapienza, Rome, Italy, mobility project (doctoral school)

## TEACHING

- Module: "Dimensionality Reduction" (15 hours), Engineering (Data major), IPF School, Montpellier.
- Module: "Spectroscopy and Hyperspectral Imaging" (70 hours), Agronomy Engineers (AgroTic), Institut Agro, Montpellier.
- Module: "Chemometrics" (42 hours), Agronomy Engineers (AgroTic specialization), Institut Agro, Montpellier.
- Module: "Scientific Approaches" (34 hours equivalent), Engineering (common track, 1st year), Institut Agro, Montpellier.
- Module: "Drones" (5 hours equivalent), SILAT Master’s program, AgroParisTech, Montpellier.

## HOBBIES

Music (Guitar, DataZombie, Logicorama), hiking, traveling, jogging, board games, fantasy and science fiction literature.

## PATENT

**Ryckewaert, M.**, Moura, D., Gobrecht, A., Roger, J.-M., Bendoula, R., Gorretta, N., & Henriot, F. (2021). System for determining a partially matched image to generate a hyperspectral image.

## AWARDS

**Best Poster Award**, IASIM Conference (2022, Esbjerg, Denmark): *Hyperspectral imaging data combined with climate data to predict stomatal conductance and transpiration of grapevine plants using Sequentially-Orthogonalized Partial-Least-Square Regression (SO-PLS)*.

**Modeling Award** at the B-CUBED Hackathon as Team Leader of an Inria/Meise team of 7 members for the project *Irokube: Deep-SDM and Critical Habitats Mapping Empowered by Data Cubes*.